

Soil water content measurements deliver reliable estimates of water fluxes: A comparative study in a beech and a spruce stand in the Tharandt forest (Saxony, Germany)

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ABSTRACT

Previous studies have shown that soil water content can vary considerably within homogeneous sites. This small-scale variability of soil water is often neglected when studying water and carbon fluxes in forest ecosystems. In this paper, the small-scale variability of soil water was analyzed at two contrasting eddy-flux sites, a Norway spruce forest and a European beech forest. Simultaneous measurements of precipitation, eddy covariance, and sap flow, from soil water content readings, were used to answer the question of how representative soil water gain is during rainfall and evapotranspiration is during dry periods.

Our study demonstrates that the spatial and temporal variability in soil water under spruce and beech was mainly due to the differences in soil properties and root intensity. This can be concluded from the fact that the pattern of soil moisture distribution and flow paths under the trees were generally stable throughout the season. As a tendency, areas with preferred accumulation of rainwater were mainly characterized by maximum soil water depletion. Therefore, the density of the installed water content sensors should correspond to the variability of soil properties as well as rooting intensity. Based on previous studies and our own results, it can be concluded that a horizontal and vertical distance between 10 and 30 cm is best suited for water content sensors to detect preferential flow paths and deliver reliable estimates of soil water balance.

Despite the occurrence of preferential flow, we found that the soil water increase during rainstorm periods and the soil water depletion during dry periods can be estimated relatively well when the small-scale variability of soil properties is considered in the experimental setup. In general, the evaporation estimates based on eddy covariance, sap flow, and soil water balance were consistent. However, compared to the spruce site, at the beech site the gap between the evapotranspiration estimates based on eddy covariance and soil water balance were often relatively large. Differences in the spatial extent of these methods can only explain these discrepancies to a certain extent. We suggest that this might be mainly due to the lack of water content sensors in the immediate vicinity of the beech tree trunk. Thus, stemflow-induced wetting and subsequent drying around the trunk could not be monitored in our study. This may result in an underestimation of evaporation from the soil under beech using the soil water balance method compared to the eddy covariance method. Finally, soil water depletion under spruce led to a significant reduction of transpiration when the actual available plant soil capacity (AWC) was <40% of the potential AWC. In contrast to the spruce stand, a reduction of transpiration of beech due to water shortage was not observed.

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1. Introduction

Soil drought is a major driver of water and carbon fluxes in forest ecosystems, whatever the tree species and climate zone be

(Granier et al., 2007). Hence, a key for the prediction of forest water and carbon budget is to understand how the soil water dynamics relate to atmospheric boundary conditions. Evapotranspiration and precipitation are the two main components of the soil water cycle in forests. From the technical point of view, precipitation at stand level can easily be measured; however, the quantification of evapotranspiration needs much more technological effort and financial resources. Many methods are currently available for the

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quantification of evapotranspiration, ranging from individual plants to soil profiles, the atmospheric surface layer, and entire watersheds (Wilson et al., 2001). Among these methods, eddy covariance and sap flow measurements have become the most popular techniques to determine forest evapotranspiration and its components. Eddy covariance measurements above the canopy provide all-year-round total evapotranspiration rates (sum of evaporation, transpiration and interception). The results represent an area up to 10^4 m^2 (Wilson et al., 2001). Compared to eddy covariance, sap flow only measures transpiration, and the spatial extent of sap flow measurements is much smaller. One advantage of the sap flow technique is that it can indicate a forest's short-term physiological response to changing environmental conditions (Köstner, 2001; Buchmann, 2002). Moreover, the combined use of sap flow and eddy covariance techniques enables a partitioning of the total evapotranspiration fluxes (McLaren et al., 2008).

Comparisons between sap flow and eddy covariance (e.g. Meiresonne et al., 2003; Wilson et al., 2001; Unsworth et al., 2004; Williams et al., 2004) or between sap flow or eddy covariance and catchment-scale water balance (e.g. Frühauf et al., 1998; Wilson et al., 2001; Ford et al., 2007; Kosugi and Katsuyama, 2007) have shown that sap flow measurements usually underestimated transpiration rates compared to other methods. These observed discrepancies were often attributed to errors associated with scaling sap flow (Williams et al., 2004; Ford et al., 2007). It is generally assumed that these errors may be due to the uncertainty in scaling-up of sap flow measurements within the tree stem and over the ecosystem (Ford et al., 2007; McLaren et al., 2008). The estimate of evaporation and transpiration from soil and understorey might be another source of error in such comparative tests. Previous studies have shown that under certain conditions up to 60% of total evapotranspiration may be caused by these processes (Roberts et al., 1980). In stands without an understorey, evaporation from soil may amount to up to 20% of total evapotranspiration (e.g. Kelliher et al., 1992; Wilson et al., 2001). Evaporation from forest soils and understorey transpiration may be determined by eddy covariance measurements below the canopy, depending on site specific turbulent conditions (e.g. Saugier et al., 1997; Unsworth et al., 2004) and lysimeter measurements (e.g. Schaap and Bouten, 1997; Silberstein et al., 2001; Deguchi et al., 2008; Müller and Bolte, 2009). Simultaneous measurements of sap flow/eddy covariance and soil water content is another method to estimate these water fluxes (e.g. Oren et al., 1998; Moore et al., 2000; Schume et al., 2005; Warren et al., 2005).

Observation of soil water dynamics is also useful for studying spatial and temporal patterns of root water uptake (e.g. Bouten et al., 1992; Schwärzel and Bohl, 2003). Furthermore, it can provide an insight into the water supply for plants during the growing season (e.g. Breda et al., 1995). Finally, soil water measurements are relevant for ecohydrological modelling because such models require information on the water content at which evaporation begins to decline (Baldocchi et al., 2004; Schwärzel et al., 2006). In all cases, monitoring soil water is linked to the inevitable problem of soil heterogeneity. Hence the question may arise: how representative are evaporation rates when calculated from soil water content readings. It is well known that soil hydraulic properties and soil water content can vary considerably within an apparently homogeneous site (e.g. Nielsen et al., 1973; Bouten et al., 1992; Deurer et al., 2000, 2001; Buttafuoco et al., 2005; Wessolek et al., 2008). Nevertheless, this small-scale variability of soil properties is often neglected when comparing eddy covariance/sap flow and soil water measurements (e.g. Wilson et al., 2001) or even when investigating the impact of soil water shortage on transpiration (e.g. Granier et al., 2007). In this context, Granier et al. (2007) called for more effort to be devoted to soil water measurements over the root zone at eddy-flux sites.

This study builds upon the conclusions and recommendations of Granier et al. (2007). It is based on combined measurements of eddy-fluxes and soil moisture dynamics at sites representing contrasting canopy situations in the Tharandt forest (Saxony, Germany). The objectives were: (1) to characterize and analyze the seasonal variation of soil water dynamics in a deciduous (beech) and a coniferous (spruce) stand with special focus on the small-scale variability in the soil; (2) to compare water fluxes estimated by the soil water balance method, sap flow technique, sap flow method, and the eddy covariance method; (3) to discuss the reliability/representativeness of soil moisture measurements taking into account the observed variability of measured soil water fluxes and information on measured precipitation, eddy covariance fluxes, and sap flow.

2. Materials and methods

2.1. Experimental sites and stand description

The study sites are located in the Tharandt forest ~20 km SW of Dresden (lower altitudes of the E. Ore Mountains, Saxony, Germany, cf. Fig. 1). This forest area has a long history of forestry documentation as well as meteorological and hydrological measurements: (i) climate data since the 1950s (ii) hydrological data of the Wernersbach catchment (4.6 km^2) since 1968 (iii) continuous flux measurements using the eddy covariance (EC) technique over a Norway spruce stand since 1996 (Grünwald and Bernhofer, 2007; Spank and Bernhofer, 2008) as well as over a European beech stand since 2005, and (iv) intensive soil hydrological measurements in the corresponding spruce and beech stands since 2005. This paper relates to the spruce site ("Tharandt Anchor Station" $50^\circ 58' \text{N}$, $13^\circ 34' \text{E}$, 380 m a.s.l.) and to the beech site ("Landberg Station" $50^\circ 59' \text{N}$, $13^\circ 29' \text{E}$, 400 m a.s.l.). The "Tharandt Anchor Station" is situated on gently sloped plateaus facing south (2°), and the "Landberg Station" on a plateau.

The Tharandt forest is characterized by a suboceanic climate. According to the long-term records of the Grillenburg weather station (1960–2007) the mean annual air temperature is 7.4°C (the maximum and minimum annual means are 9.2°C , 2000 and 5.7°C , 1996, respectively), the mean annual precipitation is 879 mm (the maximum and minimum yearly sums are 1291 mm, 1981 and 573 mm, 2003, respectively). Monthly values of the mean air temperature and the monthly sum of precipitation of the study year 2006 are contrasted with the long-term average in Table 1. The growing season in 2006 was unusually dry and warm compared to the long-term average; the climatic water balance deficit (precipitation minus grass reference evapotranspiration) was 270 mm. Also, extraordinary was the high temperatures and the low precipitation in July and September. There had not been a drier and warmer July and September since the 1960s.

The spruce stand at the Anchor Station was established by seeding in 1887. Besides the dominating spruce trees some additional tree species are present: in the within eddy covariance tower flux footprint the species of the main canopy are composed of 87% coniferous evergreen (72% *Picea abies*, 15% *Pinus sylvestris*) and 13% deciduous (10% *Larix decidua*, 1% *Betula spec.*, 2% others). The understorey mainly consists of young *Fagus sylvatica* (20%) and *Deschampsia flexuosa* (50%). The beech stand at the Landberg Station was established by wild juvenescence between 1908 and 1933. It is dominated by *F. sylvatica* (80.3%); additionally, 8.5% *P. abies*, 5.6% *P. sylvestris*, 0.5% *Betula pendula* and 0.5% *Quercus spec.* are present. Characteristics of the spruce and beech stand are presented in Table 2.

The soils of the spruce stand are loamy-skeletal podsol-brown earths (WRB: Dystric Cambisol, skeletal) and loamy podsol-brown earths (WRB: Dystric Cambisol) on rhyolite. These soils have developed from a periglacial sediment consisting of debris from

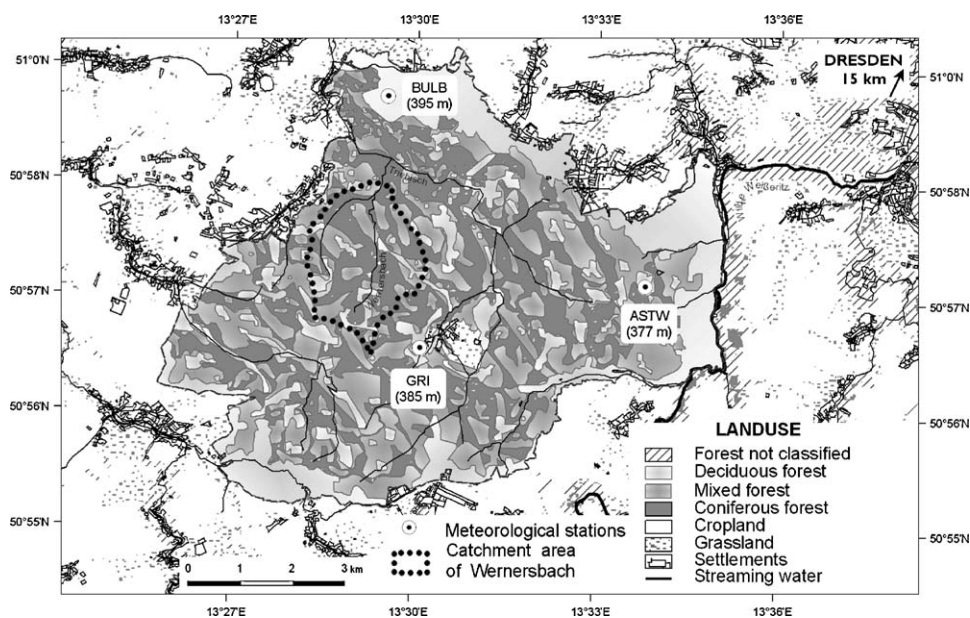


Fig. 1. Experimental sites in the Tharandt forest. ASTW = Anchor Station Tharandt, spruce stand, BULB = Landberg = beech stand, GRI = Grillenburg, grass land site.

Table 1

Monthly values of mean air temperature in 2006 and long-term average as well as monthly sum of precipitation in 2006 and long-term average. Station Grillenburg, Tharandt forest (cf. Fig. 1).

Parameter	Month												Average/sum
	1	2	3	4	5	6	7	8	9	10	11	12	
T [°C] in 2006	−4.7	−2.1	0.4	7.3	12.2	16.1	21.0	14.8	15.9	11.0	6.3	4.0	8.5
T [°C] (1960–2007)	−1.2	−0.6	2.4	6.5	11.6	14.9	16.4	15.8	12.2	8.0	3.3	−0.1	7.4
P [mm] in 2006	44	79	101	45	67	95	19	163	27	99	57	50	847
P [mm] (1960–2007)	67	60	69	66	81	83	89	97	64	60	70	74	879

rhyolite and loess. The beech forest lies mostly on loamy podsol-brown earth on rhyolite, on loamy pseudogley-brown earth on rhyolite (WRB: Endostagnic Cambisol), and loamy podsol-brown earth on sandstone. The parent material of these soils was also a solifluction deposit mixed with loess. In both forests, loess dominates the topsoil properties. Moreover, soil mapping at both sites has shown that the subsoil horizons are often compacted and characterized by mottles and related redoximorphic features.

2.2. Soil hydrological and soil physical measurements

Soil water content measurements of the spruce site were carried out using FDR (Theta probe type ML2x, Delta-T Devices, UK) on a 2.50 m long transect below the canopy of an individual spruce. A trench was dug for the horizontal installation of the FDR probes. 51 FDR probes were put in at six depths with a horizontal distance of 30 cm between them. Sensors were buried at the depths of 5, 15, 30, 50, and 70 cm each with nine replications, and at the depth of 90 cm

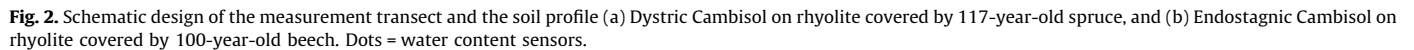
with six replications (Fig. 2a). The four needle waveguide rods ($\varnothing$ 0.3 cm) were 6 cm long, with three rods surrounding the central rod at a distance of 1 cm. Soil water content measurements of the beech site were conducted using TDR (Trase, Soil moisture equipment, USA) on a 3.10 m long transect below the canopy of an individual beech. A trench was opened for the horizontal installation of the TDR probes. 64 TDR probes were buried at six depths with a horizontal distance of 30 cm between them. Sensors were placed at the depths of 5, 15, 50, and 70 cm each with eleven replications, and at the depths of 30 and 90 cm each with ten replications (Fig. 2b). The three waveguide rods had a diameter of 0.3 cm, a length of 20 cm and a parallel distance of 2.5 cm. To minimize soil disturbance, holes for all water sensor bodies were pre-drilled into the soil profile wall, so that the soil water content measurements took place with a horizontal distance of $\sim$ 15–25 cm from the profile wall. When the equipment with sensors was fully installed, the trenches were backfilled using the original soil material.

The soil water contents were automatically recorded every 60 min. Taking measurements of soil water content in this way, as a function of time and depth, enables the calculation of the evapotranspiration (Giesel et al., 1970). This method of soil water balancing is based on the knowledge of location of the zero-flux plane (ZFP) within the soil profile. ZFP is defined as a plane which separates two zones of upward and downward movement of water in an unsaturated soil with evaporation/root water uptake and drainage occurring simultaneously (Khalil et al., 2003). Measurements of the soil pressure head allow the calculation of hydraulic gradients, so that the direction of water movement and the division between upward/downward flux can be determined.

Table 2

Characteristics of the spruce and the beech stand.

	Beech stand (2006)	Spruce stand (2004)
Tree age [years]	98	113
Canopy height [m]	30.0	29.0
Tree density [$n\ ha^{-1}$]	331	351
Tree diameter at breast height [m]	0.345	0.361
Leaf area index [$m^2\ m^{-2}$]	3.9	5.9
Above ground woody elements [$m^2\ m^{-2}$]	0.94	1.43



where P is the net precipitation rate (sum of throughfall and stemflow) [mm d^{-1}], w is the depth of the ZFP [cm], θ is the soil water content measured by FDR/TDR, and z is the vertical coordinate [cm]. It is worth mentioning that there was no understorey and ground vegetation along the TDR/FDR transects. Thus, the calculated evapotranspiration is the sum of the transpiration by the spruce/beech and evaporation from the soil. Note that the pressure heads for the determination of hydraulic

Prior to the installation of the above-mentioned sensors, the soils of the trenches were described according to the German soil classification *Ad-hoc-Arbeitsgruppe Boden* (2005). Thereafter, disturbed and undisturbed soil cores (250 cm³, with sevenfold repetition) from every horizon were taken for the lab measurement of soil texture, carbon content, bulk density, water retention curve and unsaturated hydraulic conductivity. Even though the soil texture and bulk densities of both transects seem to be quite similar, there are differences in soil hydraulic properties, mainly due to the slightly higher content of stone fragments and to the smaller amount of loess within the solum (A + B horizon) of the soil under spruce (Table 3). The outcome of this is a difference in the plant-available soil water pools between the sites. For both soil profiles, unsaturated hydraulic conductivities near saturation decreased generally with increasing depth. Conductivities of the

Soil properties of the spruce and beech site. The soil of the spruce site is a Dystric Cambisol on rhyolite, the soil of the beech site an Endostagnic Cambisol on rhyolite. C_{org} = Carbon, dB = Bulk density, AC = air capacity = volume between pF 0 and pF 2.0, AWC = plant-available water = water stored between pF 2.0 and pF 4.2, pF = log |pressure head|, n.m. = not measured.

Site	Horizon	Thickness	Sand	Silt	Clay	C _{org}	Rock fragment	dB	AC per dm	AWC per horizon	Hydraulic conductivity at pressure head [cm]			Fine roots per horizon
											−31	−63	−316	
		cm	M.-%				Vol.-%	g cm ³	Vol.-%	mm	cm d ^{−1}		roots dm ^{−2}	
Spruce	Aeh	6	17	71	12	5.83	2 to <10	1.00	9	16	0.687	0.279	0.025	15
	Bv1	24	12	73	15	1.80	10 to <25	1.15	11	48	0.273	0.125	0.018	8
	Bv2	30	11	73	16	0.76	10 to <25	1.46	4	63	0.292	0.157	0.024	4
	IISw-Bv	30	8	68	24	0.30	10 to <25	1.64	1	60	0.107	0.075	0.018	–
	IISd-Bv	10	13	68	19	0.42	10 to <25	1.55	3	19	0.344	0.185	0.029	–
	IIISd-Cv		42	47	11	0.29	50 to <75	n.m.	n.m.	n.m.	n.m.	n.m.	n.m.	–
Beech	Aeh	4	6	79	15	13.1	2 to <10	0.78	16	13	n.m.	n.m.	n.m.	50
	Bv	36	3	84	13	1.13	2 to <10	1.29	5	105	0.266	0.158	0.032	30
	Bv-Sw	35	3	82	15	0.28	2 to <10	1.64	<1	91	0.180	0.056	0.009	3
	IIBv-Sd	15	3	76	21	0.18	2 to <10	1.65	1	29	0.091	0.058	0.010	7
	IIICv		29	48	23	0.12	50 to <75	n.m.	n.m.	n.m.	n.m.	n.m.	n.m.	–

fourth horizons were up to a half order smaller than those of the top horizon. This may have been due to periglacial solifluction processes prior to the soil formation and to the higher amount of clay within these subsoil horizons. One especially noteworthy point is the very low conductivities at the depth interval from 75 to 90 cm of the soil under beech. For instance, a conductivity value of $<0.9 \text{ mm d}^{-1}$ was estimated for the pressure head of -31 cm . Table 3 clarifies that the low conductivity values correspond with higher bulk densities and lower air capacities. Root penetration of the beech as well as the spruce was hampered by soil compaction of the subsoil horizons. Approximately 90% of the fine roots of the spruce are present in the upper 25 cm of soil, whereas the maximum rooting depth of the spruce was $\sim 50 \text{ cm}$. Even though the beech had rooted down almost to 80 cm, 90% of the fine roots are concentrated in the upper 40 cm of the soil due to the low air capacity at the depth below 40 cm. Note also the differences in the root density between the spruce and the beech. Another important difference between the sites is that the soil under the spruce is characterized by wavy, irregular and broken horizons (cf. Fig. 1a). This may be due to more intensive cryoturbation during the Pleistocene.

2.3. Sap flow measurements

Sap flow was measured using Granier-style probes (Granier, 1985, 1987). Probes were 20 mm long with a diameter of 2 mm and consisted of two thermocouple junctions (copper–constantan). For measuring, each probe was inserted radially into the stem at 2 m high with a vertical distance of 150 mm. The upper sensor is heated by a heating wire (Constantan) constantly with a 0.2 W direct current (12 V, 84 mA). The reference sensor measures the temperature of the passing sap flow. Temperature differences between the two sensors were monitored in 10-s intervals, whereas 10-min averages were stored by a data logger. Two pairs of probes were installed at each sample tree (west/east). Each covers the outermost 20 mm of sapwood. Half hourly sap flow densities were calculated from observed temperature differences by Eq. (2), according to Granier (1985, 1987),

$$\text{FD} = 0.0119 \cdot \left(\frac{\Delta T_M - \Delta T_a}{\Delta T_a} \right)^{1.231} \quad (2)$$

where FD [$\text{g m}^{-2} \text{ s}^{-1}$] is the sap flow density, ΔT_M [$^{\circ}\text{C}$] the maximal temperature difference (by night or after rainfall), and ΔT_a [$^{\circ}\text{C}$] the actual measured temperature difference. To allow night-time sap flow, ΔT_M was determined over a 10-day period as suggested by Lu et al. (2004). Gap filling was performed by regression with the opposite sensor of the tree or by the sensor with highest correlation coefficient R^2 .

Sap flow at the spruce site was recorded on nine trees adjacent to the flux tower and the soil hydrological measurement plot. There was no clear dependency of sap flow density to the diameter at breast high of the measured spruce trees. On this account, stand transpiration T [mm s^{-1}] of the spruce forest was estimated by multiplying the averaged flow density of all sample trees with the cumulative sapwood area A_{scum} divided by the stand ground surface (Eq. (3)).

$$T = \text{FD}_{\text{avg}} \cdot A_{\text{scum}} \quad (3)$$

Eq. (3) requires knowledge of sapwood area A_s [cm^2] of the spruce trees of the stand. This was calculated using a functional relationship ($R^2 = 0.97$) between the tree circumferential at breast high (CBH in cm) of eight trees and the corresponding sapwood area measured using stem discs at breast height (Eq. (4)).

$$A_s = 0.158 \cdot \text{CBH}^{1.684} \quad (4)$$

Afterwards cumulative sapwood area A_{scum} [cm^2] was calculated by adding sapwood areas of all spruce trees of the stand.

At the beech site sap flow was measured on eight trees adjacent to the flux tower and the soil hydrological measurement plot. It is generally accepted that sap flow decreases radially within beech trees (Granier et al., 2000; Hölscher et al., 2005; Lüttschwager and Remus, 2007). Measurements of sap flow inside deeper sapwood are not available. Hence, results from a beech stand of similar age (95 years) (Geßler et al., 2005) were used to calculate the ratio of sap flow density from deeper xylem to the outermost 2 cm of each sample tree and to estimate sapwood area from diameter at breast high (DBH). For calculating sap flow density per tree, sap flow density of each 2 cm annulus was multiplied with the corresponding sapwood area. Afterwards all densities were summed and normalized to the related sapwood area of the tree. For up-scaling to stand transpiration, several methods are possible. In literature trees were usually divided into DBH-classes and the mean sap flow densities per class were multiply with the associated sapwood. Subsequently, to obtain canopy transpiration, all DBH-classes were summed. However, at our beech site there was no clear relationship between weighted sap flow density and diameter at breast high. As a result, the weighted mean sap flow density of all trees was multiplied with A_{scum} as described in Eq. (3).

2.4. Eddy covariance measurements

The EC flux measurements of carbon dioxide, momentum, latent and sensible heat over the Norway spruce stand (Tharandt Anchor station) were established in July 1996 on top of the experimental tower (42 m above ground) using an ultrasonic anemometer (USA; Gill Solent R2, in May 2006 replaced by a Solent Gill R3, Gill Instruments, Lymington, UK) and a closed-path infrared gas analyzer (IRGA; LI-6262 in May 2006 replaced by a LI-7000, LiCor Inc., Lincoln, NE, USA). On the tower there are measurements of additional meteorological variables (shortwave, longwave and net radiation, humidity, temperature, wind speed, carbon dioxide and water vapour concentration) available at different heights above ground. Precipitation was measured with a pluviometer (Ott, Germany) at a nearby forest opening. A throughfall trough was installed beneath the spruce canopy. This collector consists of two 10 m long and 0.16 m wide stainless steel troughs that drained into a central reservoir equipped with a water level recorder. More technical details of the flux measurements at the Anchor Station Tharandt as well as of the confidence of the derived results can be found in Grünwald and Bernhofer (2007). In 2005, a similar kind of instrumentation was established at a site stocked with European beech (Landberg Station). In contrast to the Anchor Station Tharandt, the EC system device at the Landberg Station includes an open-path gas analyzer. The USA (Campbell CSAT-3) and the IGRA (LiCor LI-7500) were installed on the top of tower at a height of 37 m, around 7 m above the canopy. A further difference to the experimental setup at the Anchor Station Tharandt was the measurement of the beech stemflow. For this purpose, three representative trees were equipped with a spiral made of polyurethane foam. Stemflow was collected in 200-l storage vessels. Recording of water levels in the vessels at 10-min intervals enabled the determination of stemflow rate.

In general, the atmospheric fluxes were calculated and corrected according to the EUROFLUX methodology (Aubinet et al., 2000). The post-processing of the fluxes includes two main corrections amongst others: (i) density correction of the LI-7500 measurements (WPL correction acc. to Webb et al., 1980) at the Landberg site and (ii) damping loss correction considered on both sites. The applied correction algorithm for the Anchor Station Tharandt is described by Spank and Bernhofer (2008). For the Landberg Station the corrections are based on Moore (1986). More

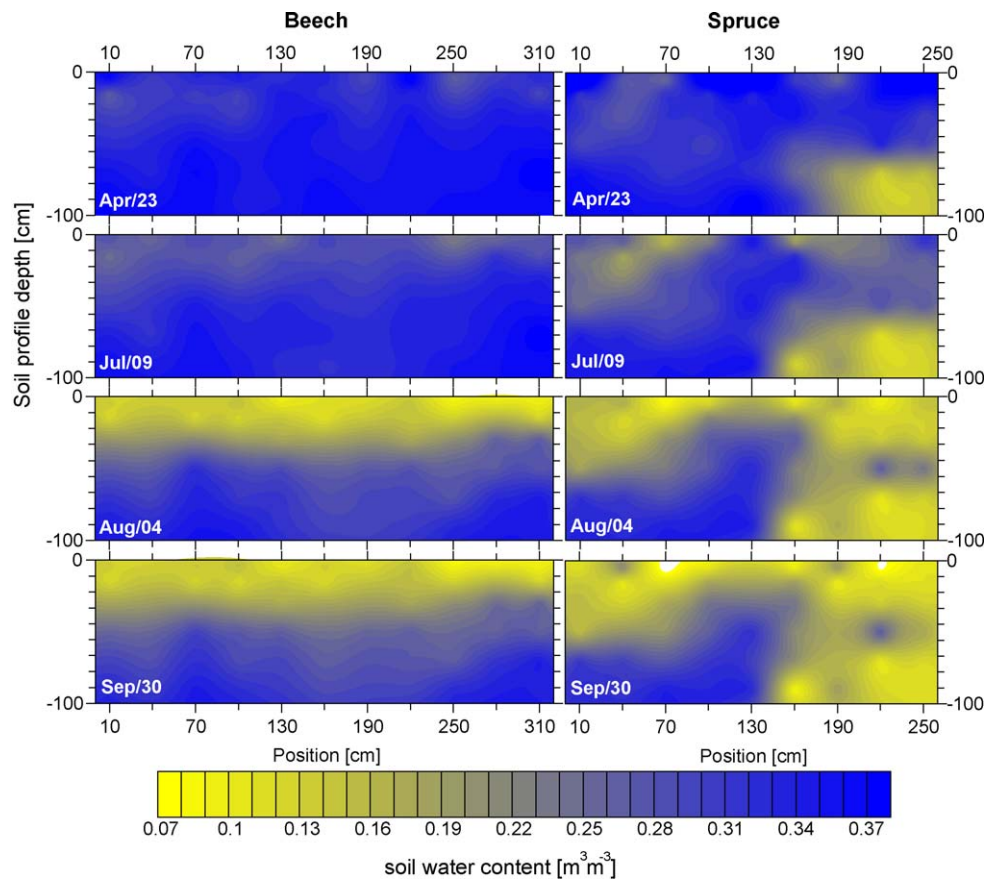


Fig. 3. Successive drying stages of the soil profiles under beech (left side) and spruce (right side) in 2006. The soil water content distribution in the beginning of growing season (bud burst on beech at April/23), in the beginning of a drought (July/9), at the end of a drought (August/4), and at the end of the main growing season (September/30). Note there was a roofing of the soil under the spruce since Jul/20.

details on the flux calculations and corrections are given in Grünwald and Bernhofer (2007) and Spank and Bernhofer (2008). In this study, we focus on the growing season in 2006. For this period, the energy balance closure (as an indicator of the quality of the half hourly flux measurements) was analyzed from the slope and the intercept of the regression between outgoing turbulent energy (sensible and latent heat flux) against available energy (sum of net radiation and heat storage changes). The slope, the intercept, and the coefficient of determination (r^2 value of the regression) of the spruce site were 0.82, 3.5, and 0.92, and the corresponding values for the beech site amounted to 0.87, 12.5, and 0.97. The closure gap is within the bounds of other studies (e.g. Wilson et al., 2002).

3. Results and discussion

3.1. Spatial and temporal variability of soil water during the season

The spatial variability of soil water under beech and under spruce is demonstrated in Fig. 3 over the main growing season in 2006. The small-scale variability of soil water was much more pronounced under spruce than under beech. The variation coefficient of the topsoil water content of the spruce-covered soil, for instance, was 29% on April 23 and even 48% on September 30. The corresponding values of the soil stocked by European beech were 8 and 19%. The spatial pattern of soil water distribution under spruce reflects the small-scale heterogeneity of soil properties well (cf. Fig. 2a). Because of long-term cryoturbation and solifluction processes during the Pleistocene periglacial, a patterned ground – stone strips and polygons – has developed at the spruce site. Due to these processes shallow ridges separated by troughs were formed.

The ridges contain a loess-enriched substrate partly mixed with debris on rhyolite (cf. Table 3); the troughs are often characterized by a relatively thin loess-enriched cover mixed with debris on stone accumulation (Fiedler et al., 1985). The soil hydrological measurement plot of the spruce site cuts through a ridge-trough-structure of this type (cf. Fig. 2a). The area with the higher stone content stands out particularly in the beginning of the growing season (between transect position 190 and 250 cm, Fig. 3). Furthermore, it can be seen from Fig. 3 that the spatial pattern of soil water distribution under spruce was persistent for the whole growing season. Finally, compared to the soil under spruce, the soil with beech desiccated more evenly during the season. This may be attributed to the more homogeneous soil profile (cf. Fig. 2b) as well as to more uniform horizontal rooting intensity.

In Fig. 4, the temporal variation of soil water storage of the root zone of the beech and spruce sites are contrasted for the growing season in 2006. Differences in the total amount of stored soil water between the sites are due to differences in the root zone thickness (cf. Table 3). The increase in soil water storage during the season was due to the infiltration of rainwater, and the decrease was due to the root water uptake by the trees. In comparison to the soil under beech, the net soil water decline under spruce was similar or greater before the full leaf expansion of the beech. Not until the leaf area index of the beech achieved its maximum did the sap flow of the beech exceed the transpiration of the spruce (cf. Table 4). This was confirmed by the soil water measurements: during the drought from July 9 to August 4, the drying of the soil profile under beech was more intensive than under spruce. At the beginning of the drought, 269 mm soil water was stored under the beech, and 203 mm water was retained at the end of this period. This corresponds to a net decline of 66 mm, or 2.4 mm d^{-1} . The soil

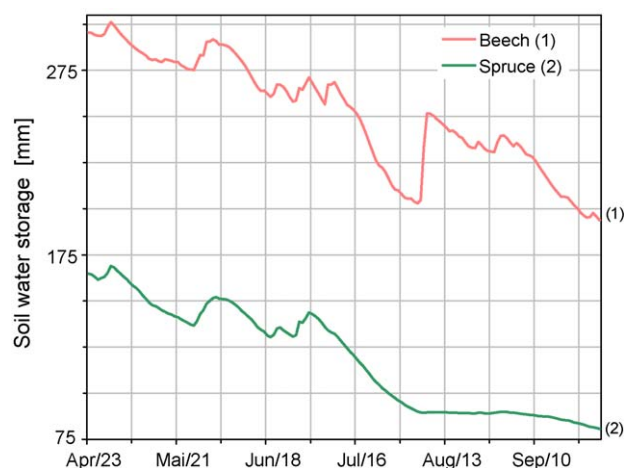


Fig. 4. Daily dynamics of soil water pools of the main root zone of beech (red line = 1) and spruce (green line = 2) during the growing season in 2006. The soil water storage of the root zone of every transect position was calculated and then averaged. Note that there was a roofing of the soil under the spruce since July/20. (For interpretation of the references to color in the citation of this figure, the reader is referred to the web version of the article.)

under the spruce declined only about 42 mm, or 1.6 mm d^{-1} during this drought. Further details are given in Section 3.3. In general, the course of the soil water storage of the sites was similar up to the beginning of August. Then, a foil cover ($4 \text{ m} \times 6 \text{ m}$) was installed above the soil stocked by spruce. The aim of rainfall exclusion was to examine if a fast lateral subsurface flux on the gently sloped plateau may be generated by heavy rainstorm events, and if such lateral water fluxes supplied the spruce in the measurement transect with additional water. According to [Germann and Zimmermann \(2005\)](#) such flow processes might be likely because of the low hydraulic conductivities of the subsoil horizon as indicated in [Table 3](#). Under the condition of such rainfall exclusion, a prompt response of the buried soil water content sensors during intensive rainfall would prove the occurrence of a fast subsurface flow. [Fig. 4](#) demonstrates that despite intensive rainfall events (e.g. 42 mm on August 6) no soil water increase was observed under the spruce during roofing (cf. [Fig. 4](#)); therefore ruling out a supply of the spruce stand by lateral subsurface fluxes.

3.2. Soil water fluxes during heavy rainstorm events

In periods with heavy rainstorm events, changes in the soil profile water content are mainly caused by the infiltration of rainwater and by runoff, although the latter was never observed at either plot in 2006. In such rainstorm periods, water loss by transpiration or by evaporation from soils can be neglected. Hence, the soil profile water increase during the rainy period should agree with the net precipitation. Balancing the soil profile water gain of individual precipitation events and comparing these results with the corresponding amount of net precipitation is one way to evaluate the reliability of soil water measurements. Determining the soil water gain during rainstorm events requires balance intervals shorter than 24 h. This may lead to errors, as shown by [Verhoef et al. \(2006\)](#) and [Fernandez-Galvez et al. \(2007\)](#), because daily variation in soil temperature may have an effect on the diurnal course of soil moisture as measured with FDR. This was due to the temperature dependence of the dielectric response of water. Similar studies on the influence of soil temperature on TDR water content measurements have been reported e.g. by [Wraith and Or \(1999\)](#) or [Olmanson and Ochsner \(2006\)](#). However, our recorded daily soil temperature fluctuations under spruce and beech were significantly smaller ($1\text{--}2 \text{ }^{\circ}\text{C}$ in 5 cm depth and $<0.5 \text{ }^{\circ}\text{C}$ in 20 cm

Table 4

Sum of evaporation measured with the eddy covariance method, with the soil water method, and the sap flow method of periods with little rainfall during the growing season in 2006. P_{gross} = gross precipitation, P_{net} = net precipitation, E = evaporation, T = stand transpiration, I = interception. The experimental setup of the measurement transect with water content sensors in a fixed spatial grid, allows to differentiate between 9 (spruce site) or 11 (beech site) vertical soil columns within the plot, each with probes at five depths. Mean = averaged over all vertical columns, max = maximal value of an individual transect position, min = minimal value of an individual transect position.

Period	Norway spruce					European beech						
	Precipitation component [mm]					Evaporation component [mm]						
	P_{gross}	P_{net}	Eddy covariance method $E + T + I$			P_{gross}	P_{net}	Eddy covariance method $E + T + I$				
			Soil water balance method $E + T$					Soil water balance method $E + T$				
			Mean	Max	Min			Mean	Max	Min		
May/01 to May/09	0.2	0.0	14.6	19.6	6.7	10.5	0.0	0.0	15.3	23.1	11.0	8.1
	5.2	2.2	28.7	31.7	12.3	19.8	4.9	3.4	47.9	42.4	24.6	18.4
	11.4	3.8	37.6	42.4	30.1	26.1	7.5	88.8	84.1	51.9	38.4	
	15.2	–	46.9	not balanced		24.3	13.8	64.1	50.8	55.5	37.1	30.2
June/02 to June/18												
July/09 to August/04												
September/01 to September/30												

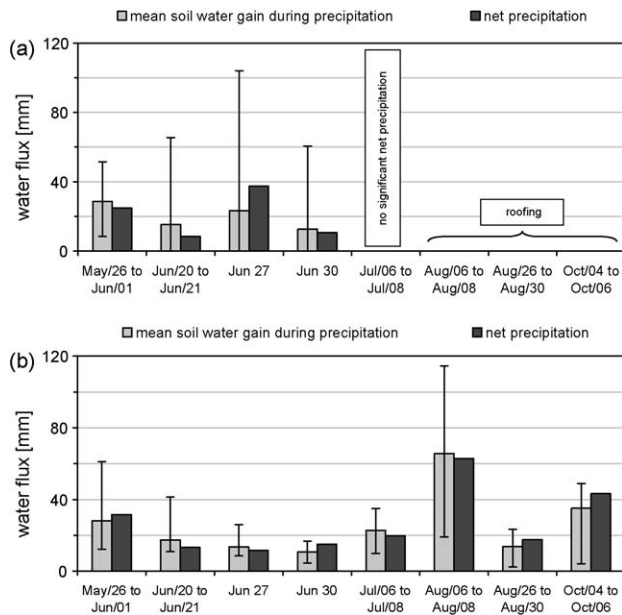


Fig. 5. Mean soil water gain during intensive precipitation events and corresponding sum of net precipitation in 2006 at the (a) spruce site and (b) beech site. The soil water gain of every precipitation event was calculated for every transect position (cf. Fig. 2) and then averaged. The error bars indicate the min and max values of individual transect positions.

depth) compared to the above-mentioned studies. Hence, possible errors in FDR/TDR water content measurements caused by soil temperature can be neglected.

The experimental setup of the measurement transect with water content sensors in a fixed spatial grid allows us to differentiate between 9 (spruce site) and 11 (beech site) vertical soil columns within the plot, each with probes at six depths. In Fig. 5 the mean soil water gain during significant precipitation events for the spruce and the beech site are compared with the corresponding net precipitation. The values were calculated for each transect position and then averaged. Overall a good agreement between precipitation and determined soil water gain was found. Discrepancies could be caused by the spatial variability of precipitation as well as by systematic errors (wind field deformation by the measuring device, evaporation of the pellicular water and evaporation from the collection vessel) in rainfall measurement. Generally, soil water increase during rainfall under spruce, as well as under beech, was limited up to the depth of 50 cm. In only a few cases of the spruce-covered soil, the water content sensors of deeper layers installed in the stone-rich horizon (between transect position 160 and 250, cf. Fig. 2) promptly responded to intense rainfall.

Furthermore, Fig. 5 also shows that the corresponding minimum and maximum values of soil water increase of individual transect positions during the precipitation events. The wide range of data indicates that preferential flow processes were predominant, particularly in the spruce-covered soil. The occurrence of preferential flow phenomena under spruce is also illustrated in Fig. 6a. Despite a net precipitation of 13 mm, large portions of this soil profile were bypassed, and the infiltrating water was preferentially accumulated at a 190 cm transect position. In contrast, Fig. 6b clarified that the wetting process under beech was more a homogeneous infiltration.

One important question for the assessment of the ecological relevance of such flow phenomena is whether the observed flow paths under the spruce were stable over a longer period or not. For this purpose, soil water storage at different transect positions of

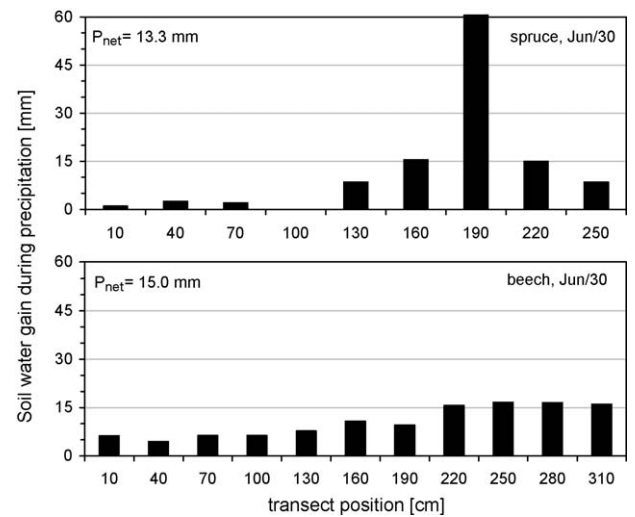


Fig. 6. Soil water gain at different transect positions during a heavy rain shower event at the spruce site (above), and the beech site (below) in June 2006.

the spruce-covered soil are presented as a time-series in Fig. 7. Three contrasting situations were chosen. At transect position 40 cm, no wetting of the soil was observed during the 18-day period despite 57 mm net precipitation. The time-series at transect position 130 cm confirms the traditional concept of piston flow, the slow increase of soil water storage due to the homogeneous infiltration of rainwater. The cumulative net soil water increase, at 46 mm, was not far off the cumulative net precipitation. The time-series at transect position 190 cm impressively illustrates the occurrence of preferential flow processes under spruce as well as

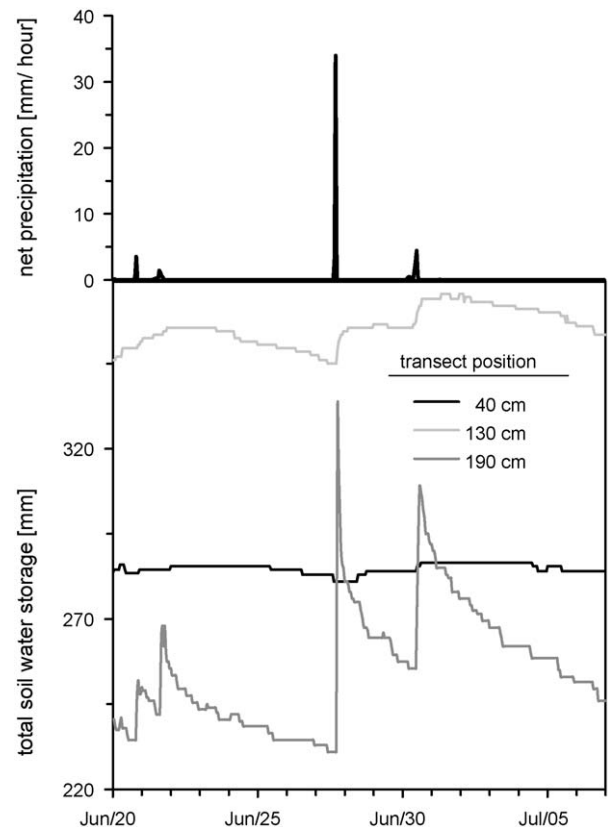


Fig. 7. Time-series of net precipitation and total soil water storage at different transect positions of the soil under spruce.

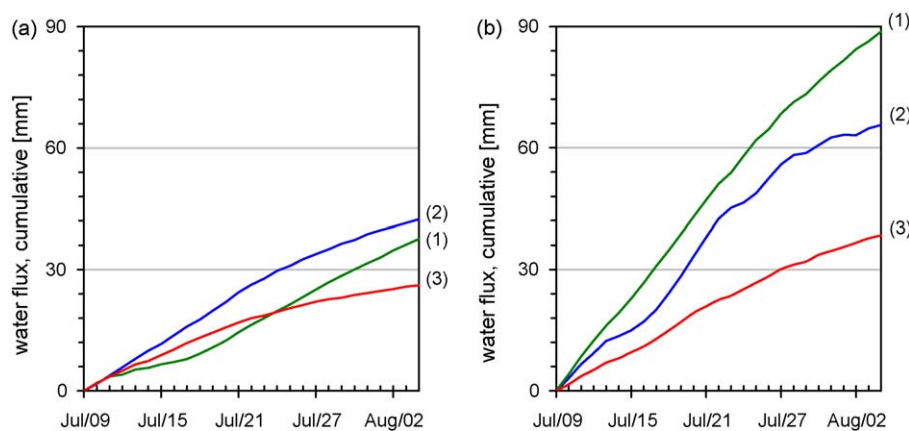


Fig. 8. Cumulative daily values of evapotranspiration measured by eddy covariance technique (green line = 1), sum of transpiration and evaporation determined by soil water measurements (blue line = 2), and transpiration measured by sap flow technique (red line = 3) for (a) the spruce site, and (b) the beech site during a drought in 2006. (For interpretation of the references to color in the citation of this figure, the reader is referred to the web version of the article.)

the stability of preferential flow path over a longer period. The soil water gain during every rainfall event always exceeded the net precipitation by many times. On June 27, for instance, the discrepancy between net precipitation and soil water gain during the rainfall event was almost 60 mm. Moreover, the FDR probes at transect position 190 cm responded very quickly to the rainfall. There was little or no delay in wetting between the topsoil and subsoil water content sensors (data not shown).

What might be the reasons for such preferential rainwater feeding of certain areas of the soil profile under spruce? Several authors (e.g. Ritsema and Dekker, 1994 or Wessolek et al., 2008) have reported that water repellency in soils may promote preferential flow. But this can be ruled out for the soil under spruce; field observations have shown that the forest floor layer was still wettable in June. What is more likely is that the preferential flow might be caused by local obstacles in the soil matrix (cf. Kung, 1990). It was already discussed in Section 3.1 that the water content under spruce always varied more than under beech (cf. Fig. 3). This may be due to the small-scale variability in horizontal rooting density as well as in soil properties. In particular, the debris is unevenly distributed within the solum (A + B horizon). In general, more debris was found over the stone accumulation. This debris may act as a barrier, channeling the water around it into preferential flow pathways. Such flow processes were described as “funnel flow” by Kung (1990). Furthermore, channeling and irregular distribution of the infiltrating water might be additionally promoted by the wavy soil surface and the wavy topsoil horizons (cf. Fig. 1a). A tracer experiment at the FDR transect would enable the percolation process to be visualized and might confirm the occurrence of “funnel flow” under spruce. We did not pursue this possibility because the FDR measurements are still going on.

3.3. Soil water fluxes during droughts and periods with low precipitation

Longer periods with little or no rainfall were chosen for the estimation of evapotranspiration by the zero-flux plane method. At the beginning of May, a zero-flux plane (ZFP) was located at the depth interval 30–40 cm within the soil profile under spruce. With the progressive drying of the soil under spruce over the season (cf. Fig. 4), the ZFP moved downward below the root zone to the depth of 60 cm in July. Contrary to the soil under spruce, no formation of a ZFP was observed under the beech between May and September; there was usually an upward movement of water from the soil profile base toward the soil surface. This may be attributed to the

very low conductivities of the subsoil horizons (cf. Table 3) and to the shape of the corresponding water retention curves, with the low water capacity ($d\theta/dh$) near saturation (curves not shown). Therefore, a slight decrease in the soil water of the subsoil horizons resulted in a large decrease in the pressure head (more negative) and induced an upward movement of water from deeper soil layers.

Periods with high evaporation demand and low precipitation are particularly suitable for cross-checking evapotranspiration estimates derived using the ZFP method (soil water balance method) against eddy covariance as well as the sap flow method. Such conditions prevailed in summer 2006 from July 9 to August 4 (cf. Table 1). In this period, the grass reference evapotranspiration (Allen et al., 2000) amounted to 154 mm, or 5.7 mm d^{-1} . During this summer drought the evapotranspiration of the beech site based on the eddy covariance was 89 mm, or 3.3 mm d^{-1} , whereas the corresponding evapotranspiration of spruce site was just 38 mm, or 1.4 mm d^{-1} . The sap flow of beech was 40 mm, or 1.5 mm d^{-1} , and of spruce 26 mm, or 1.0 mm d^{-1} during this drought. It can be seen from Fig. 8b that the soil water depletion (66 mm, or 2.4 mm d^{-1}) under beech was less than the amount of evapotranspiration measured by the eddy covariance, but more than the measured sap flow. In contrast, the soil water decrease under spruce (42 mm , or 1.4 mm d^{-1}) slightly exceeded the evapotranspiration based on eddy covariance. This might be attributed to the larger energy balance closure gap compared with the beech site between July 11 and July 21. During this period, the slope of the regression between the available energy and the sum of the turbulent heat fluxes was 0.76, and the offset 9.5. In general, the transpiration of spruce we measured are at the lower end of those known until now, compared with the transpiration values of various Central European spruce stands (e.g. Cermak and Kucera, 1987; Patzner, 2004). Note that our investigated spruce stand was significantly older than the stands in the previous studies. Therefore, an age-related decline in leaf conductance might explain the relatively low transpiration values of our spruce stand. This view might be supported by the studies of Delzon and Loustau (2005) as well as by Alsheimer et al. (1998). They found a marked decline in stand transpiration with age in coniferous forests due to decreases in both stomatal conductance and leaf area. In older trees, sapwood cell structure changes impeding the efficient transport of water to upper canopies. Therefore upper canopies are increasingly water stressed due to the hydraulic limitation of tree stems. In contrast to spruce, the measured transpiration of beech was within the range of values known from other Central European studies (e.g. Schipka et al., 2005).

In Table 4, further cross-checking of evapotranspiration estimates derived from the different methods are listed for periods with little rainfall. It was discussed above that the zero-flux plane (ZFP) under spruce was within the root zone in May and June. Accordingly, soil water decrease below the ZFP might be caused by both drainage and root water uptake. In the balance periods in May and June, we ascribed soil water decrease within the rooted parts of the soil below the ZFP to evaporation. However it may be, Fig. 9 illustrates that taking the root water uptake below the ZFP into consideration did not lead to serious errors in balancing the evapotranspiration. The contribution of the rooted layer below the ZFP to the overall evapotranspiration of the spruce was less than 10% during the balance periods in May and June. Confirmative conclusions were drawn by Cooper et al. (1990). These authors discussed the fact that root extraction from below the ZFP is negligible when balancing the evapotranspiration.

Firstly, from the results presented in Table 4, can be seen that the different methods delivered consistent results: evapotranspiration based on eddy covariance was larger than the sum of transpiration and evaporation from soil derived from the soil water balance, and the latter exceeded the transpiration measured using the sap flow method. Secondly, the evapotranspiration of the beech forest based on eddy covariance exceeded those of the spruce forest in periods with little rainfall. This was caused by the larger transpiration of beech and by the larger evaporation from the soil under the beech (difference between soil water balance and sap flow). Confirmative results from the Solling Mountains in NW Germany have been reported by Benecke and van der Ploeg (1976). In two neighbouring forests, a 125-year-old beech and 100-year-old Norway spruce stand, the complete water balance was determined. They found that in the summer months the evapotranspiration of the beech surpassed that of the spruce. Thirdly, from the results presented in Table 4, it can be seen that the evaporation from the soil under beech contributed significantly more to the total evapotranspiration, namely 31%, than the soil under spruce (22%). These differences may be attributed to the differences in the physical properties of the litter layers. Compared to beech, the litter of spruce is characterized by a very loose structure. Because of that, this layer may act as a more effective capillary barrier to reduce evaporation from soil than the litter of beech. Differences in the LAI between the spruce and the beech stand (cf. Table 2) might also explain some of the observed differences in evaporation from soil. Fourthly, in contrast to the

spruce-covered soil, eddy covariance fluxes over the beech forest were significantly larger than the corresponding evapotranspiration estimates based on soil water balance (including interception). One source of the difference between the two evapotranspiration estimates might be the fact that the tower footprint of the beech site is characterized by a larger variability of soil types compared to the spruce forest (cf. Section 2.1). Thus, discrepancies may be associated with scaling-up evapotranspiration derived from the soil water balance. A further source of the difference between the evapotranspiration estimates might be caused by an underestimation of evaporation from the soil under beech using the soil water balance method. Contrary to spruce, in beech forests the stemflow is an important precipitation component. Analyses of precipitation events of the beech forest have shown that events with precipitation >2 mm triggered stemflow, which then amounted to 16% of the rainfall. Stemflow concentrates and channels water into zones centered on the trunks of trees (e.g. Johnson and Lehmann, 2006; Liang et al., 2007). Due to our measurement setup and the lack of water content sensors in the immediate vicinity of the beech, wetting and drying around the trunk was not monitored in this study. Hence, an underestimation of evaporation from the soil under beech using the soil water balance method may explain some of the above-mentioned discrepancies between the evapotranspiration estimates.

As already discussed in Section 3.2, the experimental setup of the water content measurements enables quantification of the impact of soil water variability on the soil water budget. The enormous range in soil water gain during heavy rainstorm events was already discussed in the previous chapter. What about the variability in soil water decrease during droughts or periods with low precipitation? In addition to the mean of soil water depletion during the balance periods, Table 4 also shows the corresponding maximum and minimum values of soil water depletion of individual transect positions. In comparison to the soil water gain during heavy rainstorms (cf. Fig. 5), the variability in soil water depletion was significant lower. Interestingly, the coefficients of variation (CV) in soil water depletion decreased with increasing soil dryness. For the soil under spruce, for instance, the CV in soil water loss was 38% for the period in May and 24% for the drought in July. The corresponding CVs for the soil stocked with beech were 30% in May and 17% in July. It was demonstrated in the previous section that the preferential flow paths under spruce were stable over a longer period. Thus, we examined whether the areas with

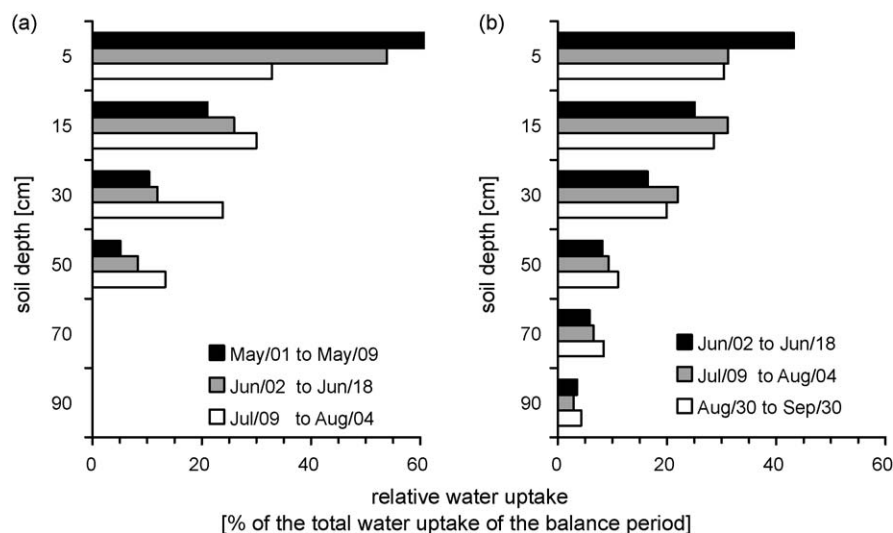


Fig. 9. Contribution of different soil layers to the total water uptake (a) by Norway spruce, and (b) by European beech at different balance periods during the growing season 2006. See also Table 4.

preferred accumulation of water during heavy rainstorm events corresponded to the areas with preferred/maximum soil water depletion during periods with little or no rainfall. As a tendency, areas which received more water during heavy rainfall were often characterized by more intensive soil water depletion compared to the corresponding mean of the respective balance period. In this context, it is noteworthy that for both soil profiles the spatial patterns of soil water depletion were, in general, stable during the growing season. In other words, maximum and minimum soil water depletion during the different balance periods was mostly observed at the same areas within the soil profile under spruce as well as beech. For the sake of completeness, it should be mentioned that the mean coefficients in variation in soil water gain range between 49 and 150% under spruce, and between 37 and 54% under beech in the examples presented in Fig. 5.

Spatio-temporal measurements of soil water are useful for studying patterns of root water uptake. The contribution of different soil layers to the total water uptake by Norway spruce and European beech is illustrated in Fig. 9. Root water uptake patterns correspond well to the root densities (cf. Table 3). For both soil profiles, soil water uptake showed a gradual downward shift as the soils became drier. During the balance period in May, for instance, more than 60% of the water which evaporated at the spruce site was taken up from the topsoil, whereas in July/August this layer only contributed to 33% of the overall evapotranspiration. During this summer drought, water was preferably taken up from the subsoil horizons despite those lower root densities compared to the topsoil. Soil water decrease below the root zone was insignificant during the balance periods (cf. also Fig. 3). Therefore it can be concluded that the capillary rise of water from deeper soil layers to the root zone did not play an important role for the water supply of the spruce stand. Also, hydraulic redistribution by tree roots does not seem to play an important role. The influence of plant-available soil water on the transpiration of spruce is presented in Fig. 10. Soil water depletion led to a significant reduction of transpiration, in particular when the ratio between the actual and potential storage of plant-available water in the rooting zone dropped below 40%. However, when the actual extractable soil water pool was >50% of the potential storage, the scatter increased and the influence of soil water storage on transpiration was less important. Then, transpiration was mainly controlled by factors such as radiation, vapor pressure deficit, or temperature.

The ability of the beech stand to send roots deeper into the soil than the spruce stand enabled the beech to bridge the summer

drought in 2006 without any significant decrease in transpiration. In contrast to the spruce stand, a reduction of transpiration of beech due to water shortage was not observed in 2006. Root water uptake patterns of beech demonstrate that the deeper soil layers significantly contributed to the overall evapotranspiration of the beech stand (Fig. 8), particularly when the soil became drier (cf. also Fig. 4). This can be also seen from the spatial distribution of soil water under beech and under spruce from the first and the last day of this summer drought (Fig. 3). Compared with the soil under spruce, a more even and in-depth desiccation was found under beech.

4. Conclusions

From the results presented in this study, it can be concluded that the soil water content sensors delivered reliable data in response to the precipitation. The detection of preferential soil water fluxes during rainfall requires a dense grid of water content sensors and measurements in high temporal resolution. Furthermore, our study showed that the increase in water pools in the soil during infiltration can be estimated relatively well – at least on average – when the small-scale variability of soil properties is considered in the experimental setup. In addition, our results confirm the recommendations of Ritsema et al. (1996) and Wessolek et al. (2008). They observed preferential flow paths between 10 and 30 cm in diameter and concluded that a horizontal and vertical distance between 10 and 30 cm is best suited for water content sensors to detect preferential flow paths.

More attention should be paid to the stemflow-induced wetting and subsequent drying around the trunk of the beech trees. Thus, we strongly recommend installing water content sensors in the immediate vicinity of the beech to prevent an underestimation of evaporation by the soil water balance method. This can be concluded from the fact that there was a relatively large gap between the evapotranspiration estimates based on eddy covariance and on recording soil water contents, which can only be explained to a certain extent by the differences in the spatial extent of both methods. Other experimenters have often found significant differences in evapotranspiration estimates between the soil water balance method and eddy covariance/sap flow technique. This may be due to inadequate numbers of soil water content sensors used in previous studies. Our study shows that a dense grid of water content sensors enable a reliable estimate of evapotranspiration, at least in periods with high evaporation demand and little precipitation. Spatio-temporal measurements of soil water in high resolution enable better observation and understanding of processes in the soil–plant–atmosphere system. In particular looking for functional relations between root water uptake, sap flow, and evaporation demand of the atmosphere or between soil water and soil temperature conditions and carbon fluxes requires the knowledge of small-scale variability of soil properties. Hence, we recommend a detailed mapping of the soils of the chosen test area before installing measurement equipment.

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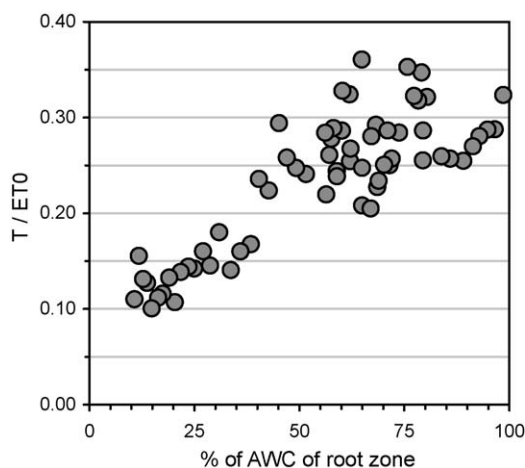


Fig. 10. Quotient of transpiration (T = sap flow) and potential evapotranspiration (ET_0 = grass reference evapotranspiration acc. to Allen et al., 2000) as a function of plant-available soil water pool (=AWC) of the root zone (0–50 cm soil depth). Norway spruce, days without precipitation, period May/01 to Aug/04 2006.

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